

BASIC CIVIL AND MECHANICAL ENGINEERING

UNIT-I

Civil engineering: Civil engineering is a professional engineering discipline that deals with the design, construction, and maintenance of the physical and naturally built environment, including public works such as roads, bridges, canals, dams, airports, sewage systems, pipelines, structural components of buildings, and railways.

Role of Civil Engineers in Society:

- Civil engineers develop and improve the services and facilities we use every day from supplying energy and clean water to homes, to processing and recycling waste, to finding solutions to problems like pollution.
- Civil engineers design and build bridges, roads, railways, and tunnels.
- Main role of Civil engineers is in surveying, planning, designing, estimation and execution of structures.
- To solve different engineering problems with the help of field experience, laboratory techniques, numerical methods, mathematical models, using computer and information technology.
- To implement management techniques for better management of man, material, machine and money.
- To carry out soil investigation for design of foundations of structures.
- To invite tenders and to select contractor for the work.
- To carry out surveying and levelling and fixing the alignments (center-line) of roads, railways, canals, tunnels, pipes etc.
- To carry out planning of buildings as per its functional needs and also has role in town and regional planning.
- To carry out the design of structures as per the principles of structural analysis and design. Civil engineer should ensure that design is safe, durable and economic.
- To supervise the work during execution and to ensure progress of work.
- To carry out valuation of land or building for the purpose of finding its sale or purchase price or taxation.
- Civil engineers has to maintain public health by providing pure water for drinking, treating waste water before disposing in to water course and to collect the solid waste of town and disposing it.
- Civil engineer has to provide basic infrastructure of the structures for projects of many other engineering disciplines, like to design machine foundations and to provide steel frame structure and sheds for industries for the mechanical engineering project.

- To construct tunnels for hydropower station, to construction cooling tower for thermal power stations and to erect transmission towers for electrical lines for electrical engineering.

Scope of Civil Engineering: Civil engineering is the art of directing the great sources of power in nature for the use and convenience of man. Basically it is concerned with planning, design and construction for environmental control, development of natural resource, buildings, transportation facilities and other structures required for health, welfare, safety, employment and pleasure of mankind. The main scope of civil engineering is planning, designing, estimating, supervising, managing and maintenance of structures like building, roads, bridges, dams etc. Broad disciplines of Civil Engineering: Civil engineering is a wide field and includes many types of structures such as residential buildings, public buildings, industrial buildings, roads, bridges, tunnels, railways, dams, canals, airports, harbours, waste water treatment plants, water supply networks and drainage networks, drainage networks etc.

Various Disciplines of Civil Engineering:

According to the type of structures and activities carried out, main branches of civil engineering are classified as follows:

- Structural Engineering
- Geotechnical Engineering
- Transportation Engineering
- Environmental Engineering
- Hydraulics and Water Resources Engineering
- Surveying and levelling

Structural Engineering:

- This discipline deals with the analysis and design of concrete and steel structures, such as multi-story buildings, bridges, towers....etc. It deals also with the study of the durability and resistibility of such structures for live loads, wind and earthquake. The study involves also the study of the properties of building materials according to the international specifications.
- Constructing Residential buildings like apartments, tenements, flats, row houses, bungalows, villas, quarters etc.
- Constructing Public buildings like schools, colleges, government offices, post offices, hospitals, shopping complexes, hostels etc.
- Constructing Industrial buildings like workshops, ware houses, stores and industrial sheds.

Geotechnical Engineering:

- This field is concerned with the study of the soil properties of the construction site and its bearing capacity. Geotechnical engineering is concerned also with the suitable solutions for any problem in the soil as well as the choice of the best and secured methods of design and construction of the foundation of engineering structures.
- Constructing several types of foundations like simple footing, well foundation, pile foundation, cofferdams and foundations of machines subjected to vibrations is the main scope of geotechnical engineering.

- It further includes constructing tunnels, earthen dams, earth work for highways and railways. It also includes soil investigation and soil testing.

Transportation Engineering:

- This discipline involves the present and future broad planning of the requirements of transportation systems, as well as the specifications of these systems such as those of roads, airports, parking lots....etc. and the management of traffic volumes with the necessary design of the traffic signals and public transportation.
- Constructing structures related to the transportation engineering like roads, railways, bridges, tunnels, ports, harbors, runways and airports.
- It also includes traffic engineering and study of highway materials.

Environmental Engineering:

- This field is concerned with the study of the necessary methods and techniques of environment protection as well as the availability of the basic life elements such as water and air with a specific level of quality to protect the mankind health and environment.
- This includes design and construction of water distribution networks, wastewater and storm water collection systems, water treatment plants and wastewater treatment for reuse in industrial and agricultural fields.
- Environmental engineering involves also the study of the different techniques of controlling air, water and soil pollution as well as the proper disposal or recycle of solid and hazardous wastes.

Water Resources and Hydraulics Engineering:

This field covers the basic concepts of water science and its related theorems and applications. This includes the methods of transporting water from sources to distribution sites through channels and pipelines, water sources and storage system, types of dams and their design methods. It involves also the study of seawater movements and shore protection.

Surveying and levelling:

- To carry out surveying for setting out of works and for preparing map of land. Leveling is carried out to measure levels and to prepare contour map. Measurements of distances and angles are taken with the help of surveying instruments. Maps are useful for planning of the construction project.

Construction Materials:

Stones, bricks, cement, lime and timber are the traditional materials used for civil engineering constructions for several centuries.

1.Stones: Stone is a ‘naturally available building material’ which has been used from the early age of civilization. It is available in the form of rocks, which is cut to required size and shape and used as building block. It has been used to construct small residential buildings to large palaces and temples all over the world.

Type of Stones: Stones used for civil engineering works may be classified in the following three ways:

- Geological
- Physical
- Chemical

Geological classification: Based on their origin of formation stones are classified into three main groups—Igneous, sedimentary and metamorphic rocks.

- (i) **Igneous Rocks:** These rocks are formed by cooling and solidifying of the rock masses from their molten magmatic condition of the material of the earth. Generally igneous rocks are strong and durable. Granite, trap and basalt are the rocks belonging to this category, Granites are formed by slow cooling of the lava under thick cover on the top. Hence they have crystalline surface. The cooling of lava at the top surface of earth results into non crystalline and glassy texture. Trap and basalt belong to this category
- (ii) **Sedimentary Rocks:** Due to weathering action of water, wind and frost existing rocks disintegrates. The disintegrated material is carried by wind and water; the water being most powerful medium. Flowing water deposits its suspended materials at some points of obstacles to its flow. These deposited layers of materials get consolidated under pressure and by heat. Chemical agents also contribute to the cementing of the deposits. The rocks thus formed are more uniform, fine grained and compact in their nature. They represent a bedded or stratified structure in general. Sand stones, lime stones, mud stones etc. belong to this class of rock.
- (iii) **Metamorphic Rocks:** Previously formed igneous and sedimentary rocks under go changes due to metamorphic action of pressure and internal heat. For example due to metamorphic action granite becomes greisses, trap and basalt change to schist and laterite, lime stone changes to marble, sand stone becomes quartzite and mud stone becomes slate.

Physical Classification: Based on the structure, the rocks may be classified as:

- (i) **Stratified Rocks:** These rocks are having layered structure. They possess planes of stratification or cleavage. They can be easily split along these planes. Sand stones, lime stones, slate etc. are the examples of this class of stones.
- (ii) **Unstratified Rocks:** These rocks are not stratified. They possess crystalline and compact grains. They cannot be split in to thin slab. Granite, trap, marble etc. are the examples of this type of rocks.
- (iii) **Foliated Rocks:** These rocks have a tendency to split along a definite direction only. The direction need not be parallel to each other as in case of stratified rocks. This type of structure is very common in case of metamorphic rocks

Chemical Classification: On the basis of their chemical composition engineers prefer to classify rocks as: Siliceous rocks, argillaceous rocks and Calcareous rocks

- (i) **Siliceous rocks:** The main content of these rocks is silica. They are hard and durable. Examples of such rocks are granite, trap, sand stones etc.
- (ii) **Argillaceous rocks:** The main constituent of these rocks is argil i.e., clay. These stones are hard and durable but they are brittle. They cannot withstand shock. Slates and laterites are examples of this type of rocks.

Tests on Stones:

To certain the required properties of stones, the following tests can conducted:

(i) Crushing strength test (ii) Water absorption test (iii) abrasion test (iv) Impact test (v) Acid test

Uses of Stones

Stones are used in the following civil engineering constructions:

- (i) Stone masonry is used for the construction of foundations, walls, columns and arches.
- (ii) Stones are used for flooring.
- (iii) Stone slabs are used as damp proof courses, lintels and even as roofing materials.
- (iv) Stones with good appearance are used for the face works of buildings. Polished marbles and granite are commonly used for face works.
- (v) Stones are used for paving of roads, footpaths and open spaces round the buildings.
- (vi) Stones are also used in the constructions of piers and abutments of bridges, dams and retaining walls.
- (vii) Crushed stones with gravel are used to provide base course for roads. When mixed with tar they form finishing coat.

2.Bricks: Brick is obtained by moulding good clay into a block, which is dried and then burnt. This is the oldest building block to replace stone. Manufacture of brick started with hand moulding, sun drying and burning in clamps. A considerable amount of technological development has taken place with better Knowledge about to properties of raw materials, better machineries and improved techniques of moulding drying and burning.

The size of the bricks are of 90 mm × 90 mm × 90 mm and 190 mm × 90 mm × 40 mm. With mortar joints, the size of these bricks are taken as 200 mm × 100 mm × 100 mm and 200 mm × 100 mm

- (i) Building Bricks: These bricks are used for the construction of walls.
- (ii) Paving Bricks: These are vitrified bricks and are used as pavers.
- (iii) Fire Bricks: These bricks are specially made to withstand furnace temperature. Silica bricks belong to this category.
- (iv) Special Bricks: These bricks are different from the commonly used building bricks with respect to their shape and the purpose for which they are made. Some of such bricks are listed below:
 - (a) Specially shaped bricks
 - (b) Facing bricks
 - (c) Perforated building bricks
 - (d) Burnt clay hollow bricks
 - (e) Sewer bricks
 - (f) Acid resistant bricks

Tests on Bricks:

The following laboratory tests may be conducted on the bricks to find their suitability:

- (i) Crushing strength
- (ii) Absorption
- (iii) Shape and size and
- (iv) Efflorescence

Classification of Bricks:

Based on their Quality the bricks used in construction are classified as:

- (i) First class bricks
- (ii) Second class bricks
- (iii) Third class bricks and
- (iv) Fourth class bricks

Uses of Bricks:

Bricks are used in the following civil works:

- (i) As building blocks.
- (ii) For lining of ovens, furnaces and chimneys.
- (iii) For protecting steel columns from fire.
- (iv) As aggregates in providing water proofing to R.C.C. roofs.
- (v) For pavers for footpaths and cycle tracks.
- (vi) For lining sewer lines.

3. CEMENT: Cement is a commonly used binding material in the construction. The cement is obtained by burning a mixture of calcarious (calcium) and argillaceous (clay) material at a very high temperature and then grinding the clinker so produced to a fine powder. It was first produced by a mason Joseph Aspdin in England in 1924. He patented it as Portland cement.

Types of Cement:

In addition to ordinary portland cement there are many varieties of cement. Important varieties are briefly explained below:

- (i) White Cement
- (ii) Coloured Cement
- (iii) Quick Setting Cement
- (iv) Rapid Hardening Cement
- (v) Low Heat Cement
- (vi) Pozzulana Cement
- (vii) Expanding Cement

- (viii) High Alumina Cement
- (ix) Blast Furnace Cement
- (x) Acid Resistant Cement
- (xi) Sulphate Resistant Cement
- (xii) Fly Ash Blended Cement

Properties of Ordinary Portland cement:

Chemical properties: Portland cement consists of the following chemical compounds:

- | | |
|--|-----|
| (a) Tricalcium silicate $3 \text{ CaO} \cdot \text{SiO}_2$ (C3S) | 40% |
| (b) Dicalcium silicate $2\text{CaO} \cdot \text{SiO}_2$ (C2S) | 30 |
| (c) Tricalcium aluminate $3\text{CaO} \cdot \text{Al}_2\text{O}_3$ (C3A) | 11 |
| (d) Tetracalcium aluminate $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$ | 11 |

Physical properties or Testing's on cement: The following physical properties should be checked before selecting a portland cement for the civil engineering works. IS 269–1967 specifies the method of testing and prescribes the limits:

(a) Fineness (b) Setting time (c) Soundness (d) Crushing strength.

(a) Fineness: It is measured in terms of percentage of weight retained after sieving the cement through 90 micron sieve or by surface area of cement in square centimeters per gramme of cement. According to IS code specification weight retained on the sieve should not be more than 10 per cent. In terms of specific surface should not be less than $2250 \text{ cm}^2/\text{gm}$.

b) Setting time: A period of 30 minutes as minimum setting time for initial setting and a maximum period of 600 minutes as maximum setting time is specified by IS code, provided the tests are conducted as per the procedure prescribed by IS 269-1967.

(c) Soundness: Once the concrete has hardened it is necessary to ensure that no volumetric changes takes place. The cement is said to be unsound, if it exhibits volumetric instability after hardening. IS code recommends test with Le Chatelier mould for testing this property. At the end of the test, the indicator of Le Chatelier mould should not expand by more than 10 mm.

(d) Crushing strength: For this mortar cubes are made with standard sand and tested in compression testing machine as per the specification of IS code. The minimum strength specified is 16 N/mm^2 after 3 days and 22 N/mm^2 after 7 days of curing.

Uses of Cement:

Cement is used widely for the construction of various structures. Some of them are listed below:

- (i) Cement slurry is used for filling cracks in concrete structures.
- (ii) Cement mortar is used for masonry work, plastering and pointing.
- iii) Cement concrete is used for the construction of various structures like buildings, bridges, water tanks, tunnels, docks, etc.
- (iv) Cement is used to manufacture lamp posts, telephone posts, railway sleepers, piles etc.
- (v) For manufacturing cement pipes, garden seats, dust bins, flower pots etc. cement is commonly used.
- (vi) It is useful for the construction of roads, footpaths, courts for various sports etc.

4. Sand: Sand is a natural product which is obtained as river sand, nalla sand and pit sand. However sea sand should not be used for the following reasons: It contains salt and hence

structure will remain damp. The mortar is affected by efflorescence and blisters appear. It contains shells and other organic matter, which decompose after some time, reducing the life of the mortar. Sand may be obtained artificially by crushing hard stones. Usually artificial sand is obtained as a by-product while crushing stones to get jelly (coarse aggregate)

Sand is used in mortar and concrete for the following purpose:

- It sub-divides the paste of binding material into thin films and allows it to adhere and spread.
- It fills up the gap between the building blocks and spreads the binding material.
- It adds to the density of the mortar.
- It prevents the shrinkage of the cementing material.
- It allows carbon dioxide from the atmosphere to reach some depth and thereby improve setting power.
- The cost of cementing material per unit volume is reduced as this low cost material increases the volume of mortar.
- Silica of sand contributes to formation of silicates resulting into the hardened mass.

The properties of good sand are:

- It should be chemically inert. It should be free from organic or vegetable matter.
- It should be free from salt.
- It should contain sharp, angular and coarse grains.
- It should be well graded. It should be hard.

5. PLAIN CONCRETE: Plain concrete, commonly known as concrete, is an intimate mixture of binding material, fine aggregate, coarse aggregate and water. This can be easily moulded to desired shape and size before it loses plasticity and hardens. Plain concrete is strong in compression but very weak in tension. The tensile property is introduced in concrete by inducting different materials and this attempt has given rise to RCC, RBC, PSC, FRC, cellular concrete and Ferro cement. In this chapter proportioning, mixing, curing, properties, tests and uses of plain concrete is dealt in detail. The other improved versions of concrete are explained and their special properties and uses are pointed out.

Major ingredients of concrete are:

1. Binding material (like cement, lime, polymer)
2. Fine aggregate (sand)
3. Coarse aggregates (crushed stone, jelly)
4. Water.

A small quantity of admixtures like air entraining agents, water proofing agents, workability agents etc. may also be added to impart special properties to the plain concrete mixture.

Depending upon the proportion of ingredient, strength of concrete varies. It is possible to determine the proportion of the ingredients for a particular strength by mix design procedure.

Functions of Various Ingredients:

Cement is the binding material. After addition of water it hydrates and binds aggregates and the surrounding surfaces like stone and bricks. Generally richer mix (with more cement) gives more strength. Setting time starts after 30 minutes and ends after 6 hours. Hence concrete should be laid in its mould before 30 minutes of mixing of water and should not be subjected to any external forces till final setting takes place.

Coarse aggregate consists of crushed stones. It should be well graded and the stones should be of igneous origin. They should be clean, sharp, angular and hard. They give mass to the concrete and prevent shrinkage of cement. Fine aggregate consists of river sand. It prevents shrinkage of cement. When surrounded by cement it gains mobility enters the voids in coarse aggregates and binding of ingredients takes place. It adds density to concrete, since it fills the voids. Denser the concrete higher is its strength.

Water used for making concrete should be clean. It activates the hydration of cement and forms plastic mass. As it sets completely concrete becomes hard mass. Water gives workability to concrete which means water makes it possible to mix the concrete with ease and place it in final position. More the water better is the workability.

However excess water reduces the strength of concrete. To achieve required workability and at the same time good strength a water cement ratio of 0.4 to 0.45 is used, in case of machine mixing and water cement ratio of 0.5 to 0.6 is used for hand mixing.

Tests on Concrete:

The following are some of the important tests conducted on concrete:

1. Slump test.
2. Compaction factor test.
3. Crushing strength test

Slump Test: This test is conducted to determine the workability of concrete. It needs a slump cone for test. Slump cone is a vessel in the shape of a frustum of a cone with diameter at bottom 200 mm and 50 mm at top and 300 mm high. This cone is kept over a impervious platform and is filled with concrete in four layers.

Each layer is tamped with a 16 mm pointed rod for 25 times. After filling completely the cone is gently pulled up. The decrease in the height of the concrete is called slump. Higher the slump, more workable is the concrete. The desired values of slumps for various works have been shown in fig 1.1.

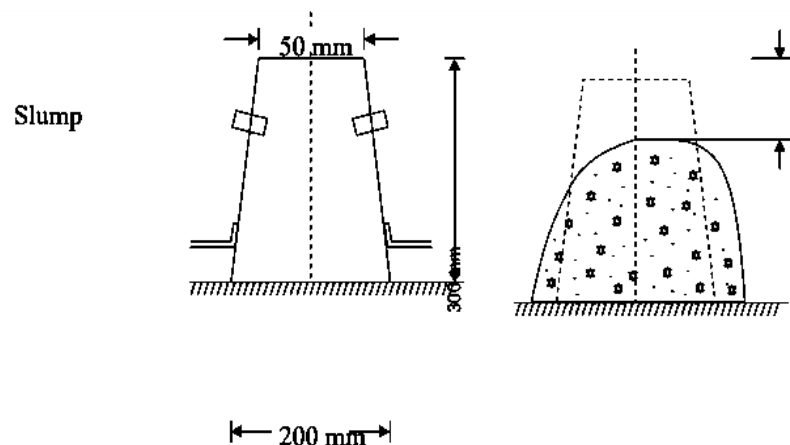


Fig 1.1 Slump Test

Compaction Factor Test: This is another test to identify the workability of concrete. This test is conducted in the laboratory. The test equipment consists of two hoppers and

a cylinder fixed to a stand, the dimensions and the distances between the three vessels being standardized. Vessel A and B are having hinged bottoms whereas cylinder C is having fixed bottom.

Top vessel A is filled with the concrete to be tested. As soon as it is filled, the hinged door is opened. Concrete is collected in vessel B. Then the hinged door of B is opened to collect concrete in cylinder C. The concrete in cylinder C is weighted. Let it be W1. Now cylinder is again filled with the sample of concrete in 50 mm layers, which is compacted by ramming and vibrating. Then the weight of compacted concrete is determined. Let this weight be W2. The ratio $W1 / W2$ is termed as compaction factor.

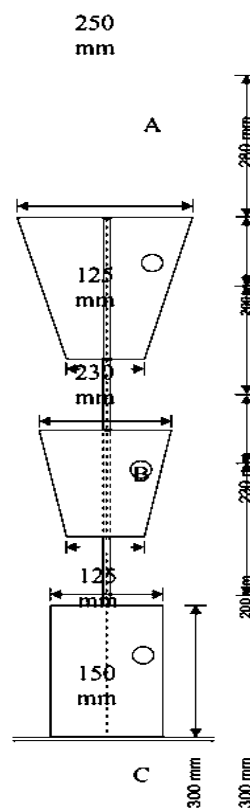


Fig. 1.2 Compaction factor

Crushing Strength Test: Metallic moulds of size 150 mm × 150 mm × 150 mm are used for casting concrete cubes. Before filling mould, it is properly oiled on its inner surfaces, so that cubes can be easily separated. Fresh cube is filled with concrete to be tested in 3 layers and kept in the room. After 24 hours, cube is removed from the mould and kept under water for curing. After 28 days of curing cubes are tested in the compression testing machine. In this test cubes are placed over the smooth surface which is in contact with side plates of mould.

5. FERROUS METALS: A ferrous material is the one in which iron is a main constituent. Iron ore is first converted into pig iron and then pig iron is subjected to various metallurgical

processes to mix different percentage of carbon and to get the following three useful ferrous materials:

1. Cast iron—carbon content 1.7% to 4.5%
2. Wrought iron—carbon content 0.05% to 0.15%
3. Steel—carbon content 0.25% to 0.25%.

All ferrous materials contain about 0.5 to 3% silica, less than 2% manganese, 0.15% sulphur and 0.6% phosphorous.

1. Cast Iron: Important properties of cast iron are:

- (a) Compression strength is 700 N/mm² and tensile strength is 150 N/mm².
- (b) It is brittle and does not absorb shocks
- (c) Its specific gravity is 7.5.
- (d) Its structure is coarse, crystalline and fibrous.
- (e) It cannot be magnetised.
- (f) It does not rust-easily.
- (g) It has low melting point of about 1200°C.

Uses of Cast Iron:

1. It is used for making rain water and sanitary pipes, sanitary fittings and manhole covers.
2. It is used for making railings and spiral stair cases.
3. Fire gratings; cover for pumps and motors and brackets are made with cast irons.

Uses of steel in the construction:

Steel is a durable and recyclable material perfect for building residential properties. Construction workers use this material in a variety of ways within their industry. Here are some of the common uses of steel in the construction industry.

1. Structural Sections
2. Reinforcing Bars
3. Sheet Products
4. Non-Structural Appliances
5. Transportation Networks

Tests performed on steel:

1. **Tensile strength test:** The tensile strength test is one of the most common tests used to evaluate the quality of a material. This test determines the maximum load that a material can withstand before it breaks or deforms permanently.
2. **Chemical Composition Analysis:** The chemical composition of the steel reinforcement bars can also affect their quality. A chemical composition analysis can determine the percentage of elements such as carbon, manganese, and sulphur in steel.

3. **Bending test:** The quality of the steel is determined by whether or not it cracks or fractures during this process. The bend test is an effective way to measure the strength and durability of steel, as it provides a visual representation of the steel's ability to withstand stress and pressure.
4. **Visual Inspection:** This is a simple yet effective way to identify any signs of rust, cracks, or deformities in the bars.
5. **Ultrasonic Test:** Ultrasonic testing is a non-destructive method of testing the quality of materials that uses high-frequency sound waves to detect internal defects or irregularities. This method is particularly useful for detecting tiny cracks, voids, or other flaws in solid objects that are invisible to the naked eye.
6. **Corrosion Test:** Corrosion testing is another way to test the quality of steel reinforcement bars. This test involves exposing them to a corrosive environment and then checking for signs of corrosion.

Factors to be considered in Building Planning:

Basic requirements of a building:

The planning and construction of a building should be aimed at fulfilling the following requirements:

1. Strength and stability
2. Dimensional stability
3. Resistance to dampness
4. Resistance to fire
5. Heat insulation
6. Sound insulation
7. Protection against termite attack
8. Durability
9. Security against burglary
10. Lighting and ventilation
11. Comforts and convenience
12. Economy

PLANNING: All buildings should be properly planned, keeping in view the various requirements of a good building. Except strength requirement, all other requirements of a good buildings are taken care at the stage of planning. Strength requirement is taken care during structural design of building components. However in planning the building by-laws of the statutory authorities should not be violated. Planning of the building is an art combined with science.

Principles of planning of buildings may be grouped into:

1. Orientation
2. Energy efficiency
3. Utility
4. Other requirements of the building.

The factors we have to consider while designing site layout:

- Construction sequence.
- Quantity of materials to be stored.
- Parking of workers.
- Sanitary facilities.
- Soil conditions.
- Safety.
- Site Accessibility.
- Information Signs.

Nature of Buildings: 'The nature of building' will draw a line connecting some of our (HARQUITECTES) works, focussing on the logic and consequences of construction systems, bioclimatic strategies, structural compositions and more or perhaps how we organise matter to foster things to happen naturally once again.

Typical Layouts of a Residential Building- Industrial Building- Commercial Building:

Residential Construction

Residential construction refers to the sector of construction that deals with building structures meant for dwelling purposes. These include single-family homes, condominiums, apartments, townhouses, and other types of housing units.

One key feature of residential construction is its focus on aesthetics and comfort. Architects and engineers working on residential projects consider factors such as lighting, space usage, and interior design to create a comfortable and aesthetically pleasing living space.

Residential construction also tends to involve smaller teams of workers and shorter construction timelines, as the projects are often smaller in scale compared to commercial and industrial construction projects. These projects generally involve standard materials such as wood, bricks, and concrete, and they follow local building codes and regulations to ensure the safety and sustainability of the housing units.

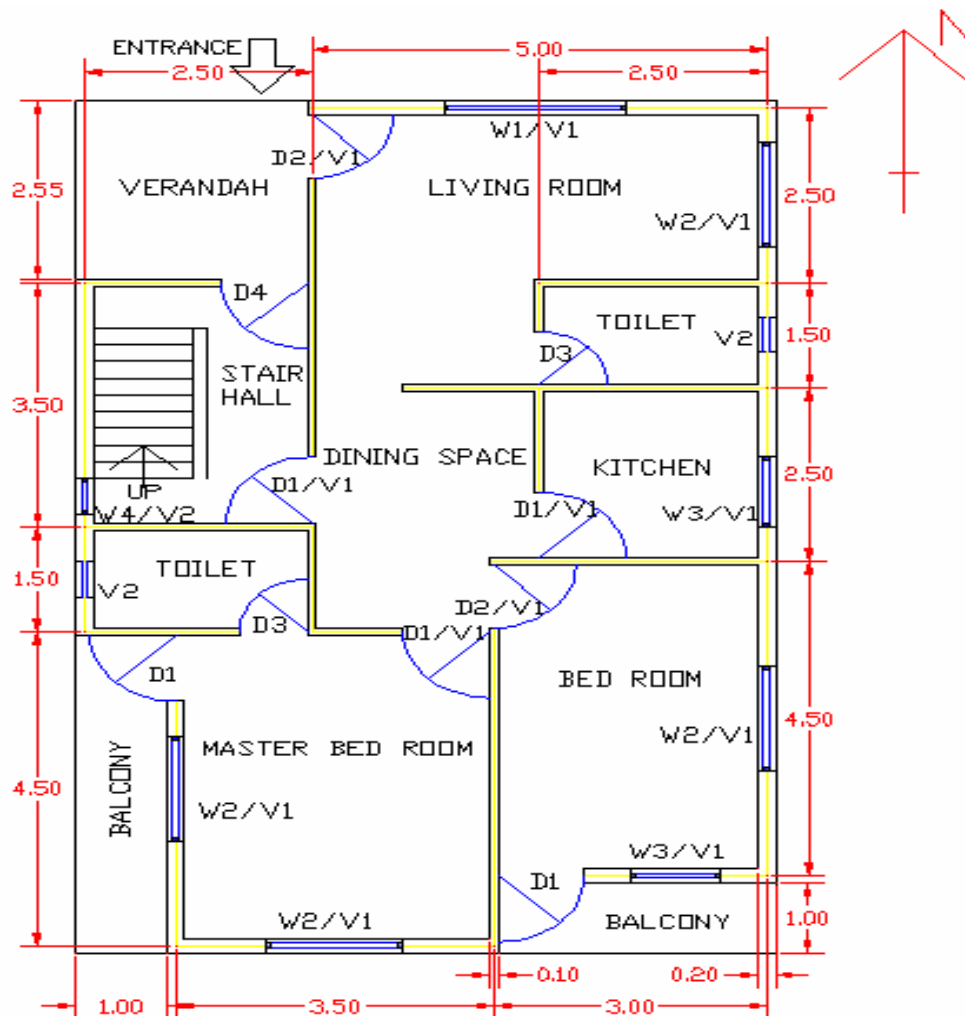


Fig. Lay out of a Residential Building

Industrial Construction: Industrial construction refers to the building of facilities intended for industrial activities. These include factories, power plants, warehouses, and other facilities where goods are produced or distributed.

Industrial construction is often characterized by large-scale projects that require highly specialized knowledge and expertise. Given the technical complexity and high stakes involved (think chemical plants or nuclear power stations), industrial construction often demands the most stringent standards and regulations of all construction types.

In addition to the usual building materials like steel and concrete, industrial construction projects may also require specialized materials to meet the unique needs of the industrial facility, such as resistant materials for high-temperature environments or specific safety measures for hazardous material handling.

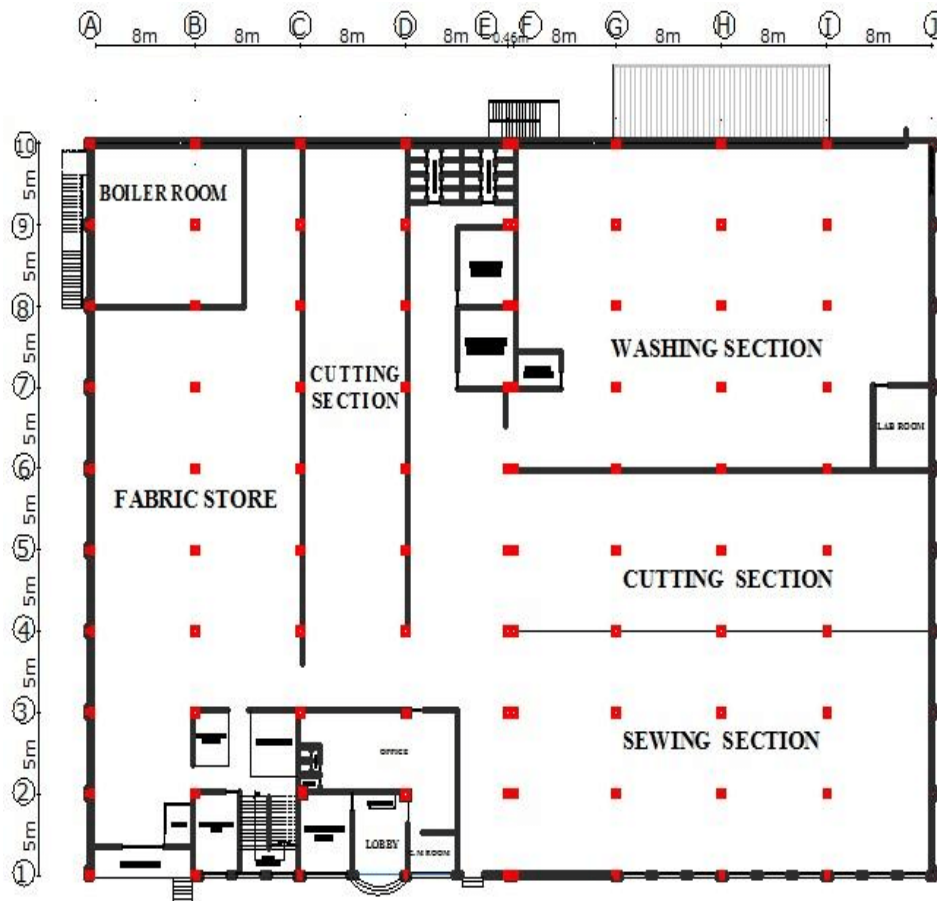


Fig. Layout of a Industrial Building

Commercial Construction:

Commercial construction involves creating structures for business purposes. These structures include office buildings, shopping malls, restaurants, hotels, hospitals, and other similar facilities.

Unlike residential construction, commercial projects are typically larger in scale and involve more complex designs and systems. For instance, commercial buildings often require more sophisticated HVAC (Heating, Ventilation, and Air Conditioning) systems, elevators, and large-scale electrical systems.

Furthermore, commercial construction projects must adhere to stringent building codes and zoning regulations, with a particular emphasis on safety and accessibility. It's also not uncommon for these projects to have larger budgets, longer timelines, and larger teams of workers compared to residential construction.

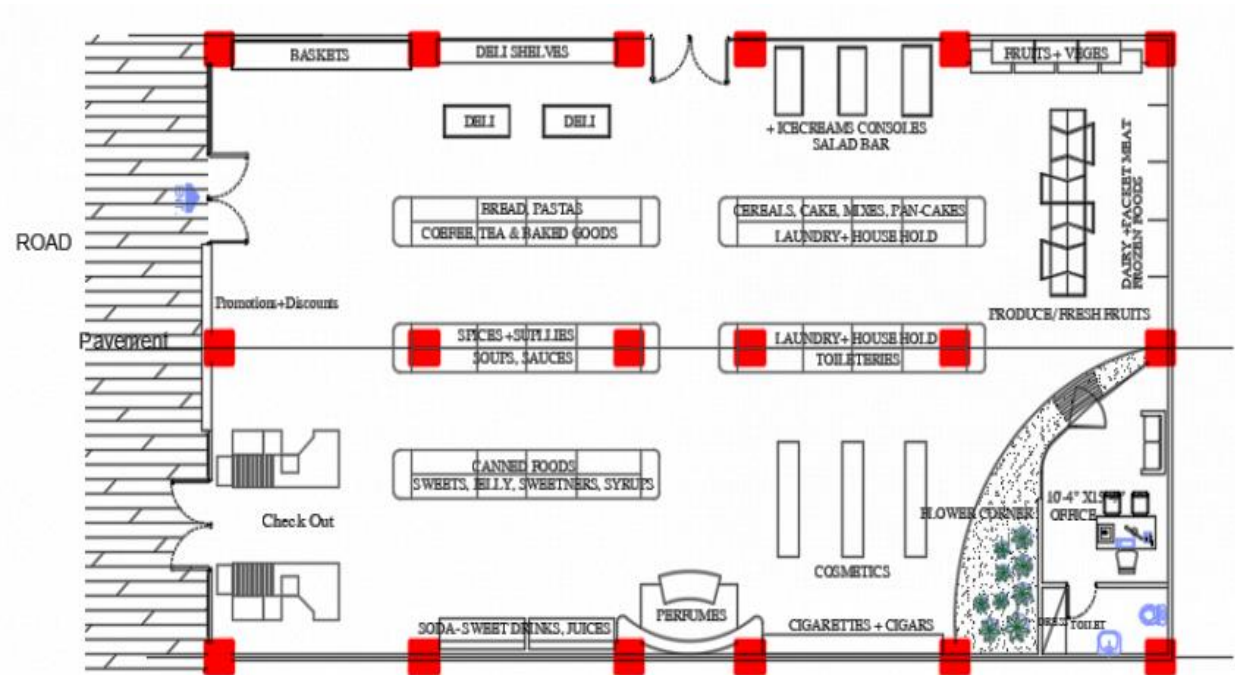


Fig. Layout of a Commercial Building (Supermarket)